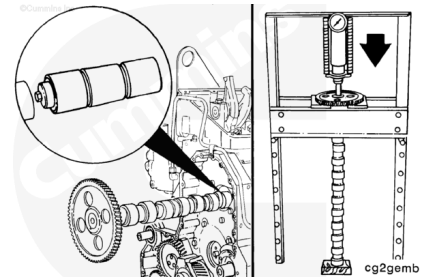


Trainings ▾

Remove

⚠ CAUTION ⚠

Support the camshaft gear on the center hub when pressing it off the camshaft. Failure to do so can result in damage to the gear.

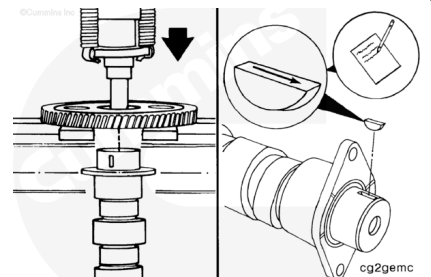


Remove the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/35/35-001-008.html>).

Install the camshaft and gear assembly in a hydraulic press.

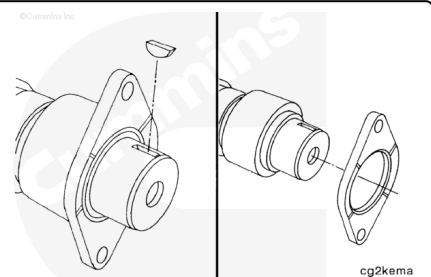
Press the camshaft from the gear.

Note : If the camshaft key is marked with an arrow, record the direction the arrow is pointed.



Remove the camshaft key.

Remove the thrust plate.



Install

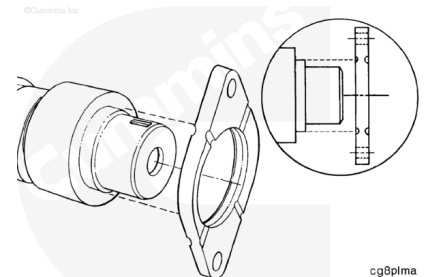
The accompanying chart lists different camshaft key part numbers, the degree of offset, and the approximate injector timing change from nominal.

Camshaft Keys

Key Part No.	Degree of Offset (To the Camshaft)	Change in Push Rod Travel	
		mm	in.
3009953	0.00	0.000	0.0000
3030893	0.25	0.051	0.0020
3009948	0.50	0.102	0.0040
3030894	0.75	0.152	0.0060
3009949	1.00	0.203	0.0080
3030895	1.25	0.254	0.0100
3009950	1.50	0.305	0.0120
3030896	1.75	0.356	0.0140
3009951	2.00	0.406	0.0160
3030897	2.25	0.457	0.0180
3030898	2.50	0.508	0.0200

IT200nb

Install the thrust plate on the camshaft.

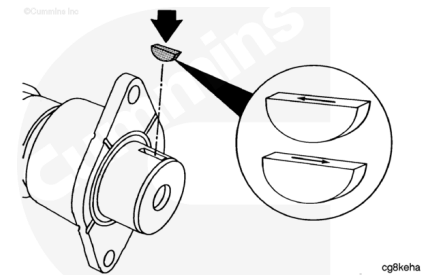


cg8plma

Be sure the gear mounting surface is clean and free of oil, dirt, or debris.

Use a leather hammer to install the camshaft gear key.

Note : If the same camshaft and gear are used again, use the same part number key as the one that was removed. Be sure the arrow on the key is pointing in the same direction as when it was removed. If a different camshaft and gear are used, install a straight key.

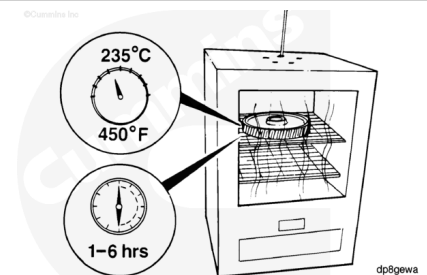


cg8keha

⚠ WARNING ⚠

Use insulated gloves, Part Number 3823730, and/or hot clamp pliers, Part Number 3823732, when handling heated parts. Hot parts can cause serious personal injury.

Heat the gear in an oven for a minimum of 1 hour, but no longer than 6 hours.



dp8gewa

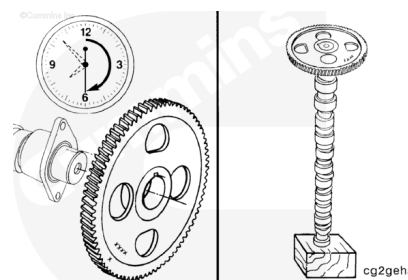
Measurements

	celsius	fahrenheit
Temperature	235	450

Use Lubriplate™ 105, or equivalent, to coat the camshaft nose before installing the camshaft gear.

⚠ CAUTION ⚠

The timing marks and gear part number must be facing away from the camshaft when the gear is installed to prevent engine damage.



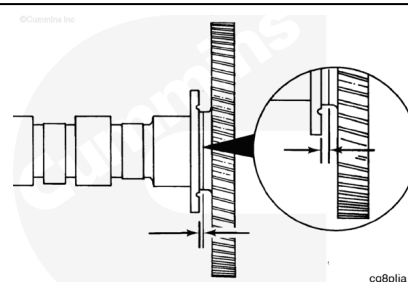
The gear **must** be installed within 30 seconds after being removed from the oven.

Remove the gear from the oven. Align the keyway in the gear with the key in the camshaft, and install the gear on the camshaft.

Keep the camshaft in a vertical position with the gear up until the gear has cooled.

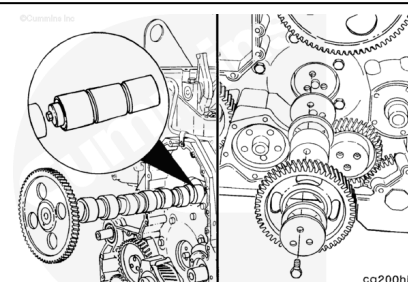
Measure the clearance between the gear hub and the thrust plate.

mm		in
0.13	MIN	0.005
0.33	MAX	0.013



Install the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/35/35-001-008.html>).

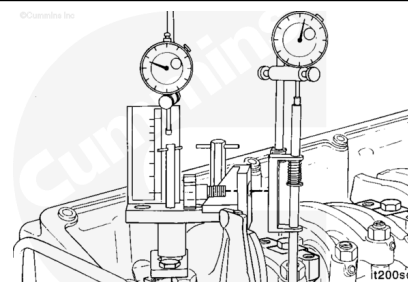
Install the camshaft idler gear. Refer to Procedure 001-036 (</qs3/pubsys2/xml/en/procedures/35/35-001-036.html>).



⚠ CAUTION ⚠

Failure to check engine timing can cause severe engine damage.

Check the engine timing. Refer to Procedure 006-025 (</qs3/pubsys2/xml/en/procedures/35/35-006-025.html>).



Install the gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/35/35-001-031.html>).

Operate the engine to normal operating temperature and check for leaks.

